Predicting Average Lake Mixed-Layer Temperature Using Multiple Linear Regression

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Introduction and Motivation

Lake temperature is an important environmental variable because it reflects how heat is stored, transferred, and mixed within a lake. In this project, we focus on average lake mixed-layer temperature, which is the temperature of the upper portion of the lake that is actively mixed. This layer matters because it is directly related to lake-atmosphere interactions and can affect seasonal mixing, dissolved oxygen levels, algae growth, and broader aquatic ecosystems.

Our research question is: Can we predict average lake mixed-layer temperature using atmospheric and lake conditions from Copernicus climate data? The response variable is average lake mixed-layer temperature, and the predictors are two-meter temperature, total cloud cover, lake bottom temperature, and surface solar radiation. We chose these four predictors because they all represent different physical phenomena that control lake temperature. Two-meter temperature measures the temperature two meters above the lake surface, cloud cover tells us how cloudy the sky is, lake bottom temperature indicates how warm the bottom of the lake is, and surface solar radiation determines how much solar radiation reaches the surface of the lake.

Our research question matters because lake mixed-layer temperature is related to seasonal lake behavior and the timing of warming and cooling periods. In particular, the winter cooling period can reset the thermal memory of the lake, meaning that the lake's temperature at the end of winter strongly affects how quickly and how much the lake warms in spring, setting the tone for the rest of the year. For this reason, our analysis focuses on winter period observations from January and December, when atmospheric cooling and stored lake heat are especially important.

The goal of our analysis is to test how well a multiple linear regression model can explain and predict average mixed-layer temperature from a set of physically meaningful predictors.

Dataset Description

The data used in this project came from ERA5-Land, downloaded as a GRIB file from the Copernicus Climate Data Store. ERA5-Land is a gridded climate dataset, so the original data were stored as raster layers rather than a spreadsheet or data frame. Each layer is a spatial grid over our selected region for one variable at one time step.

We selected the Lake Superior area as our study region, with spatial coverage of longitude -92.5 to -84.0 and latitude 46.0 to 49.2. So we are analyzing a regional grid rather than a single point location, where each raster layer contains many grid cell values across the region.

Although the basic layer names identified 2-meter temperature, total cloud cover, and surface solar radiation directly, two of the layers initially appeared as "undefined" in the GRIB layer names. After looking into the GRIB metadata, we identified var10 of table 228 from ECMWF as lake bottom temperature and var8 of table 228 from ECMWF as lake mixed-layer temperature.

For units, lake mixed-layer temperature, lake bottom temperature, and 2-meter temperature were measured in Kelvin. Total cloud cover was measured on a 0 to 1 scale, where 0 is clear conditions and 1 is complete cloud cover. Finally, surface solar radiation was measured in W*s/m², equivalent to Joules per square meter.

The extracted time index contained 173 timestamps per variable. The timestamps were January 1 and December 1, beginning in 1940 and continuing through January 2026. Because we only included two winter-period timestamps per year in the data (one per month), our analysis should be interpreted as a winter-period prediction model rather than a full annual model.

Preprocessing and Regression Dataset

The first preprocessing step was to load the ERA5-Land and GRIB file into R using the terra package. After loading the file, we inspected the first few layers to understand how the variables were stored. We saw that the first five layers followed the following order: 2-meter temperature, total cloud cover, lake bottom temperature, lake mixed-layer temperature, and surface solar radiation. This same five-variable pattern repeated through all of our 865 layers.

Since the variables repeated every five layers, we separated them using sequence indexing. Layers 1, 6, 11, … were extracted as 2-meter temperature, layers 2, 7, 12, … as total cloud cover, layers 3, 8, 13, … as lake bottom temperature, layers 4, 9, 14, … as lake mixed-layer temperature, and finally layers 5, 10, 15, … as surface solar radiation.

Each extracted layer was still a raster map, not one single value (Figure 1). And since our goal was to fit a standard multiple linear regression model with one variable per time step, we converted each raster layer into one regional average. For each variable, we averaged all the grid cells over our study region using the global mean function in the terra package. This gave us one mean value per variable per time step. After this spatial averaging, we put all the variables into one regression dataframe. Each row represented one time step, and each column represented a response variable / predictor.

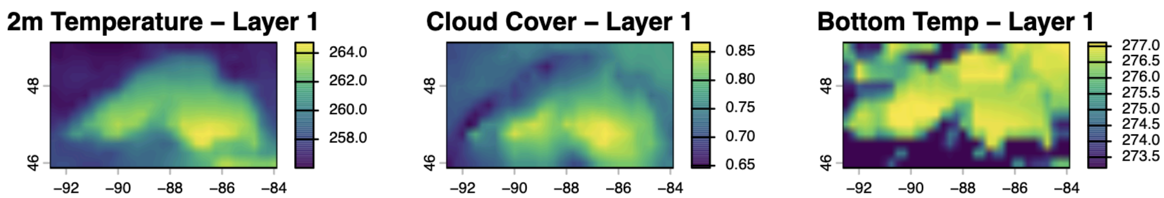

Figure 1) First-Layer Raster Plots

We also checked for missing values before we started modeling. There were none, so no observations were removed. The final dataset contained 173 complete observations and was ready for multiple linear regression.

Methods and Assumptions

But before fitting the regression models, we first looked at the relationships between the variables (Figure 2). From the correlation analysis, one interesting relationship was between cloud cover and solar radiation, which had a correlation of about -0.51. This makes physical sense because more cloud cover usually means less incoming solar radiation, so we decided to test their interaction later.

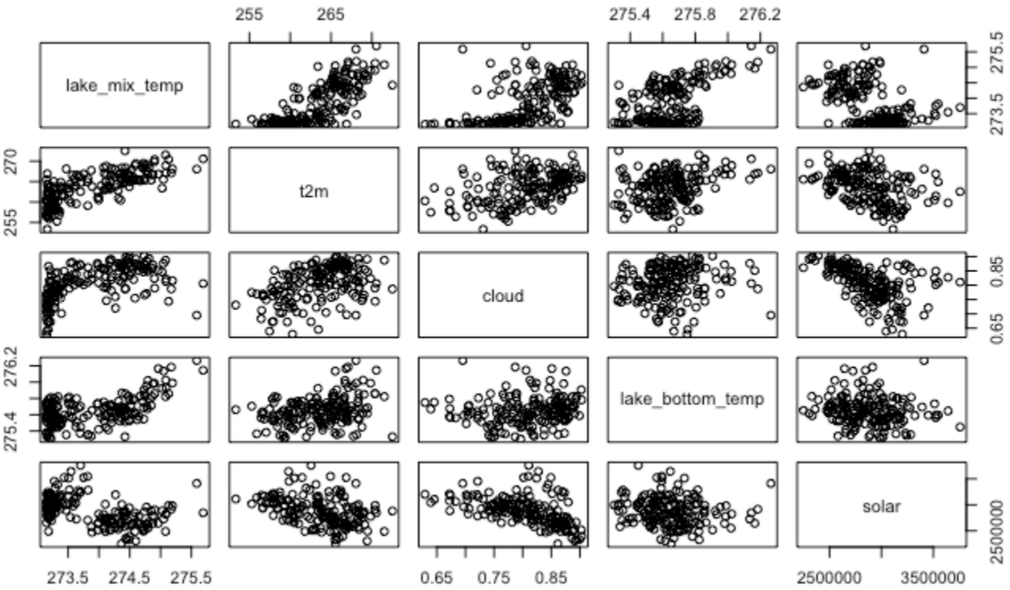

Figure 2) Pairs Plot of all 5 variables

For model fitting, we started with a baseline model that included all 4 predictors. The baseline model found all variables to be significant at the 0.05 level, with two-meter-temp (t2m) as the most significant and solar as the least significant predictor. The adjusted R^2 of this initial model was 0.7526.

After fitting the full model, we compared it against simpler models to check whether a simpler structure could perform almost as well. The atmospheric-only model tested whether weather-related variables alone were enough, and the two-temperature model tested whether the main temperature variables alone were enough.

We used AIC and adjusted R^2 for the comparison (Figure 3). For AIC, smaller is better because it penalizes models that add extra terms. For adjusted R^2, larger is better because it accounts for how many predictors are used. Both simpler models had higher AIC and lower adjusted R^2 than the full model, so the full model was a better starting point. We then added the cloud:solar interaction because cloud cover can change how solar radiation affects lake temperature. This gave the lowest AIC and highest adjusted R^2 among these early models.

Model	Predictors	AIC	Adj_R2
<chr>	<chr>	<dbl>	<dbl>
Full model	t2m + cloud + lake_bottom_temp + solar	113.50	0.753
Atmospheric-only model	t2m + cloud + solar	181.77	0.631
Two-temperature model	t2m + lake_bottom_temp	153.91	0.684
Full model + cloud:solar interaction	t2m + cloud + lake_bottom_temp + solar + cloud:solar	107.40	0.763

4 rows

Figure 3) Comparison of Models with Different Number of Predictors

From our full baseline model, we tried variable selection to see if we could simplify the model before moving forward. The three variable selection techniques that we tried were Stepwise AIC (both forward and backward), Stepwise BIC, and Lasso Regression. Stepwise AIC and BIC both kept the full model, and Lasso did not shrink the solar coefficient to zero, even though it got relatively close. The consistent results of these variable selection techniques indicate that all four predictors meaningfully add to predictive power and are worth keeping in the model.

Then, we checked for multicollinearity by calculating VIF (variance inflation factor). All VIF values were well below 5, confirming no multicollinearity concerns (Figure 4).

```
     t2m            cloud lake_bottom_temp            solar
1.397635         1.473144         1.112270         1.411373
```

Figure 4) VIF Values

Even though the VIF values do not pass the threshold for concern, interactions are still possible and are highly plausible for environmental variables. This is why we decided to investigate interactions. We tested for all interactions by creating a full interaction model, which revealed that the interactions t2m:lake_bottom_temp (p-value = 4.56e-07) and cloud:solar (p-value = 0.0453) were significant. We then added these two interactions to the full model to create a refined model. This model had a lowered AIC of 84.94 and an adjusted R^2 of 0.79.

We believed that we could make this model better by adding quadratic (squared) terms. We created fitted scatterplots of each predictor against lake mixed temperature to see which ones could follow a quadratic pattern (Figure 5). Based on these scatterplots, it looks like t2m and lake_bottom_temp could possibly be better explained with a quadratic term. We then tested for all quadratic terms by creating a full quadratic term model, which revealed that the two-meter temperature squared I($t2m^2$) variable is highly significant (p-value = 0.0126). We added this term to our refined model, which led to an adjusted R^2 of 0.806 and AIC of 73.66, making this our most accurate model yet.

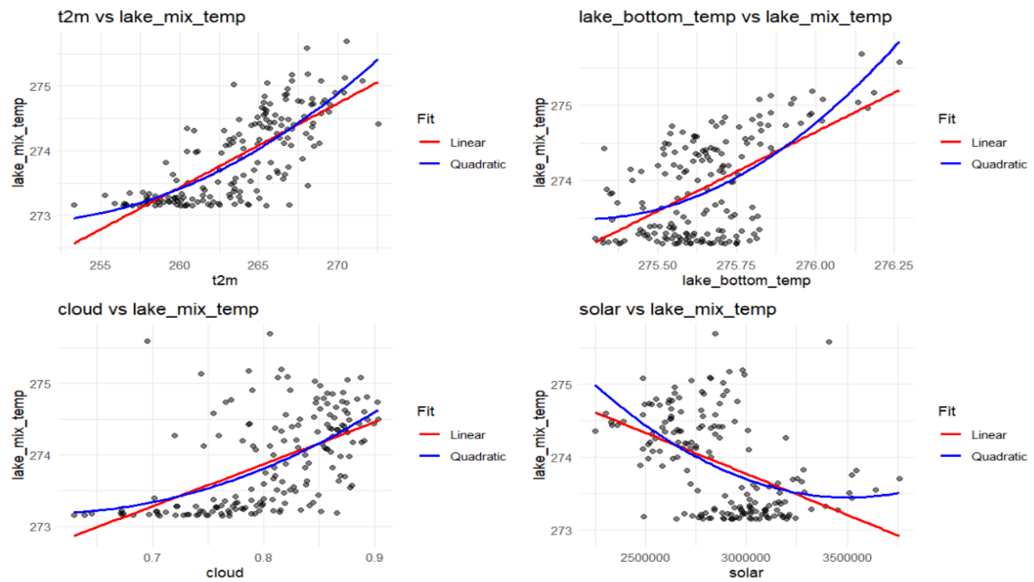

Figure 5) Fitted Scatterplots of Predictor Variables Against Lake Mixed Temperature

At this point, we were quite pleased with the model performance and wanted to see whether or not it meets the normality and constant variance assumptions. Based on the diagnostic plots shown below, the Residuals vs Fitted plot has a mild downward trend at higher fitted values, suggesting slight residual non-linearity. Q-Q plot shows heavy tails, indicating nonlinearity, which is a cause for concern (Figure 6). The Scale-Location plot shows mild heteroscedasticity at lower fitted values, and the Residuals vs Leverage plot shows no observations beyond Cook's distance.

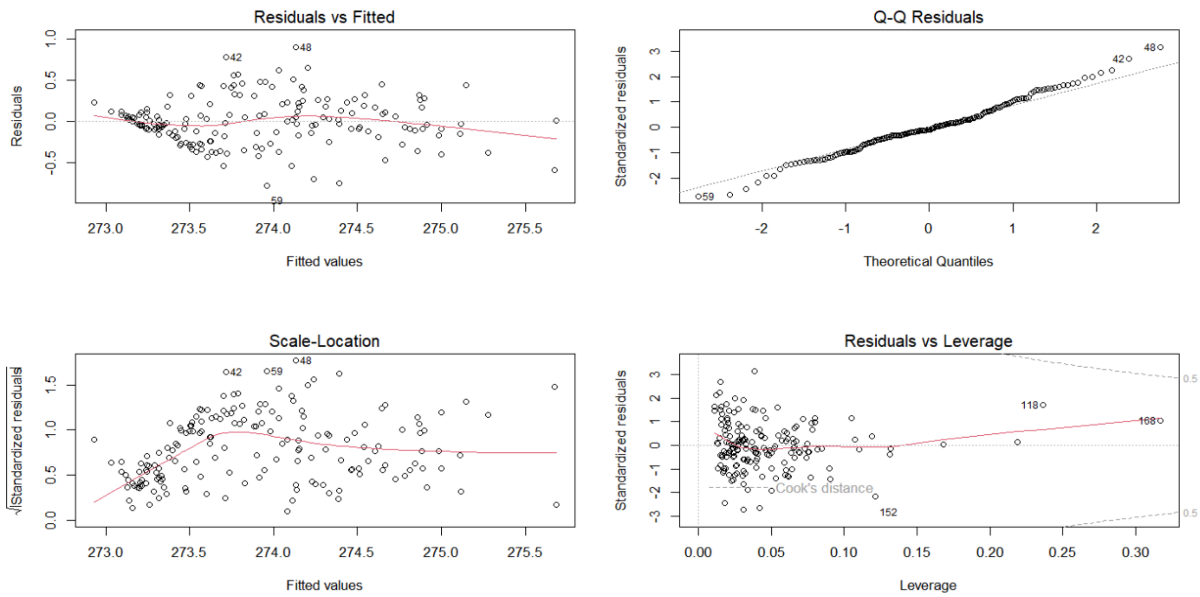

Figure 6) Diagnostic Plots for Refined Quadratic Model

We conducted further tests, including the Breusch-Pagan test for heteroscedasticity, Shapiro-Wilk normality test, and the Durbin-Watson test for autocorrelation. The

Breusch-Pagan test had a p-value of 0.021, and since this is less than 0.05, it means that we reject the null hypothesis of homoscedasticity. This means there is statistically significant evidence that the residual variance is not constant. The Shapiro-Wilk normality test had W = 0.98 and a p-value of 0.2351. Since p is greater than 0.05, we fail to reject the null hypothesis of normality, meaning that there is no significant evidence of non-normality in your residuals.

The implication of not meeting the constant variance assumption is that our OLS standard errors are likely biased, which makes hypothesis tests and confidence intervals unreliable. Although the coefficient estimates themselves are still unbiased, inference is compromised.

For this reason, we decided to try transforming the data in hopes that it might fix the violation of the constant variance assumption. We tried the anomaly transformation, a data transformation where you subtract the mean of a variable from each observation, leaving only the deviations from the mean. When we fit our model we saw not much of an improvement in diagnostics, a higher RSE (0.30) and smaller R^2 than before, which led us to try a different method of transformation.

We also tried the Box-Cox transformation, giving an optimal lambda of -2, which is at the extreme negative end, signaling that the outcome has a severe right skew. When looking at the diagnostic plots, they hadn't improved much, so we decided not to go with this option since it would add minimal predictive power to the model but lose a lot of interpretability.

Finally, we investigated outliers by creating an influence plot sen in Figure 7. The influence plot showed several outliers of interest, one of them being point 118, which can be seen to have both a large Cook's Distance (shown by the dark blue color) as well as large hat value (above 0.10 threshold). Upon further investigation of the values of this point, it can be found that lake bottom temperature is at its maximum value.

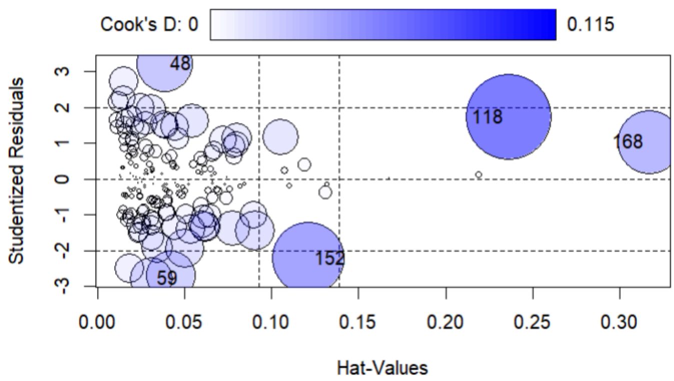

Figure 7) Influence Plot

Therefore, for our final model, we decided to remove this outlier. This resulted in the smallest adjusted AIC of 71.201 and adjusted R^2 0.80 and RSE of 0.2892 (Figure 8).

```
Call:
lm(formula = lake_mix_temp ~ t2m + I(t2m^2) + cloud + lake_bottom_temp +
    solar + t2m:lake_bottom_temp + cloud:solar, data = model_data[-118,
    ])

Residuals:
     Min       1Q   Median       3Q      Max
-0.78908 -0.15502 -0.01314  0.17468  0.89609

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)           1.090e+04  2.544e+03   4.285 3.11e-05 ***
t2m                  -4.273e+01  9.549e+00  -4.475 1.43e-05 ***
I(t2m^2)              5.259e-03  1.399e-03   3.759 0.000237 ***
cloud                 1.428e+01  5.077e+00   2.812 0.005521 **
lake_bottom_temp     -3.735e+01  9.284e+00  -4.023 8.76e-05 ***
solar                 2.876e-06  1.442e-06   1.995 0.047748 *
t2m:lake_bottom_temp  1.453e-01  3.501e-02   4.150 5.33e-05 ***
cloud:solar          -3.794e-06  1.719e-06  -2.208 0.028662 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2892 on 164 degrees of freedom
Multiple R-squared:  0.8108,	Adjusted R-squared:  0.8027
F-statistic: 100.4 on 7 and 164 DF,  p-value: < 2.2e-16
```

Figure 8) Summary of Final Model with Observation 118 removed

Out-of-Sample Validation

To estimate the true predictive skill of our model, we opted to use grouped 10-fold cross-validation. This cross-validation approach was chosen based on reasons related to the features of the data we were analyzing.

Firstly, our model contains a nonlinear term ($t2m^2$) and two interaction terms (cloud:solar and t2m:lake_bottom_temp), both of which are prone to causing overfitting in a model. Using a 10-fold cross-validation would allow for a stronger penalization of overfitting because this would allow for enough data to be removed at a time (10%) without overestimating the true error rate.

Secondly, the presence of a strong outlier in Observation 118 means that there may be more borderline outliers present in the data. A 10-fold cross-validation would smooth the impact of an individual anomaly by averaging the model performance across larger, balanced validation chunks.

Thirdly, due to our team focusing on the winter months of December and January in the data, these months naturally have a dependency on each other due to them being so close together in time and season. This means that these months contain a lot of information about each other, causing data leakage if these months are split in the training/predicting process that would inflate the predictive skill and lead to an overly optimistic error rate. For this reason specifically, we opted for a grouped 10-fold design that keeps consecutive months together in the folding process so that the model is trained on 90% of the years and tested on a completely unseen 10% of the years.

As for our process, we based our resampling pipeline on n = 172 observations, leaving out the Observation 118 outlier. To create the appropriate grouping, we created a new “year” column in the dataset that assigned consecutive rows to the same seasonal bucket by subtracting an observation’s time index by 1, using integer division to divide the difference by 2, and then discarding the remainder so that consecutive months of different calendar years remain together. We then used this as the grouping with k = 10 folds to keep these months on the same side of the training or predicting process.

Using the caret package in R, we trained the model on 90% of historical years and predicted 10% of block years, the model executed across all folds using the OLS method.

Our results were highly satisfactory compared to our in-sample R^2, the out-of-sample R^2 remaining strong at 0.8012 (Figure 9). This means that even when looking at entirely new winter seasons with no information from neighboring months to predict off of, our model is able to explain over 80% of the variance in lake mixed-layer temperatures.

Additionally, our error metrics suggest our model can predict very precisely, the root mean squared error equaling 0.2846 and the mean absolute error equaling 0.2220. These low values indicate that our model’s error predictions are, on average, only off by less than a third degree Kelvin, demonstrating that our model is stable and rarely makes large mispredictions.

```
Linear Regression

172 samples
  4 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 140, 146, 150, 165, 154, 160, ...
Resampling results:

  RMSE       Rsquared   MAE
  0.2846081  0.8011672  0.2219699

Tuning parameter 'intercept' was held constant at a value of TRUE
```

Figure 9: Results of the Grouped 10-Fold Cross-Validation

Discussion and Conclusion

Overall, our model proved to be highly disciplined as a predictive framework for tracking the thermal dynamics of Lake Superior by incorporating atmospheric and lake variables from the Copernicus Climate Data Store. Through a rigorous model development process that examined a variety of variable and interaction combinations, checked for multicollinearity, and targeted outlier mitigation, we successfully established a stable multiple linear regression structure. Our model directly aligns with environmental physics by capturing the nonlinear curvature of air temperature along with critical air-lake and cloud-solar thermodynamic interactions. We evaluated the mathematical validity of this framework through the implementation of a grouped 10-fold cross-validation strategy that prevented temporal data leakage and further proved the strength of our model’s ability to explain the variance in lake mixed-layer temperature across decades of unseen historical data with an out-of-sample R^2 of 0.8012, MAE of 0.2220°K, and RMSE of 0.2846°K.

Though we can confidently conclude that mid-winter lake mixed-layer temperatures are highly predictable using this set of environmental variables across multiple decades, it cannot be concluded that this relationship implies absolute physical causation since our model maps statistical associations across these relationships. Furthermore, this model cannot be used year-round as we only accounted for winter data across all variables. These variables that we optimized for the December and January months should not be generalized to explain lake thermodynamics during significant periods such as spring thaws that may have other, more significant variables that are specific to that time of year.

The primary limitation of this study is the seasonal restriction and temporal autocorrelation present in continuous climate time-series records. Though we mitigated month-to-month data leakage within singular seasons in our out-of-sample validation, the OLS assumption of completely independent errors cannot be fully prevented due to thermal memory from preceding seasons across nearly a century of data.

As for future prospects that can help us go beyond our current model and dataset limitations, an initial action that we can take is expanding our available data. By pulling from data of global climate drivers like El Niño Southern Oscillation (ENSO), Arctic Oscillation (AO), or Pacific Decadal Oscillation (PDO), we could capture multi-decadal climate cycles that can broaden the scope and prediction capabilities of our model. Another change we could make is using individual pixel and grid-cell data modeling rather than averaging over a full layer. This could lead to more precise and spatially explicit representations of specific areas of Lake Superior, leading to better predictability for more focused locations. Finally, we could employ machine learning architectures through models such as Random Forests to reduce manual computational labor and further detect higher-order interactions, which would allow us to understand more comprehensively all of the influences on lake mixed-layer temperature.

References

Copernicus Climate Change Service, Climate Data Store, (2023): ERA5 hourly data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). DOI: 10.24381/cds.adbb2d47 https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means?tab=download